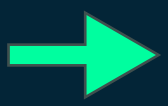
Top SQL Interview Questions



Question

How to find duplicates in a given table?

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

Query

```sql
SELECT emp_id,
COUNT(emp_id) AS records
FROM emp
GROUP BY emp_id
HAVING COUNT(emp_id)>1;
```

emp_id	records
1	2

Question

What is difference between UNION & UNION ALL?

Suppose these two same tables named as emp & emp1

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

Fetched manager_id through UNION ALL, it brought every record including duplicates.

```sql
SELECT manager_id
FROM emp
UNION ALL
SELECT manager_id
FROM emp1;
```

Result Grid

manager_id
4
5
4
2
6
2
2
2
2
4
5
4
2
6
2
2
2
2

While UNION only fetched records without duplicates.

```
SELECT manager_id
FROM emp
UNION
SELECT manager_id
FROM emp1;
```

manager_id
4
5
2
6

Question

Employees who are not present in department table?

emp

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

dept

dep_id	dep_name
100	Analytics
300	IT

6

Query

Employees who aren’t in dept table

```
SELECT * FROM emp
WHERE department_id NOT IN (SELECT dep_id FROM dept);
-- but this approach is not good for query performance
```

emp_id	emp_name	department_id	salary	manager_id
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

As subquery isn’t good for query performance, let’s try differently.

```sql
SELECT e.*, d.*
FROM emp e
LEFT JOIN dept d
ON e.department_id = d.dep_id;
```

Using LEFT JOIN it output NULL for those employees who aren’t in dept table

emp_id	emp_name	department_id	salary	manager_id	dep_id	dep_name
1	Ankit	100	10000	4	100	Analytics
2	Mohit	100	15000	5	100	Analytics
3	Vikas	100	10000	4	100	Analytics
4	Rohit	100	5000	2	100	Analytics
5	Udit	200	12000	6	NULL	NULL
6	Agam	200	12000	2	NULL	NULL
7	Sanjay	200	9000	2	NULL	NULL
8	Ashish	200	5000	2	NULL	NULL
1	Saurabh	900	12000	2	NULL	NULL

Now filter those records having NULL in dept table

```sql
SELECT e.*, d.*
FROM emp e
LEFT JOIN dept d
ON e.department_id = d.dep_id
WHERE d.dep_name IS NULL;
```

Following employees aren’t in department table.

emp_id	emp_name	department_id	salary	manager_id	dep_id	dep_name
5	Udit	200	12000	6	NULL	NULL
6	Agam	200	12000	2	NULL	NULL
7	Sanjay	200	9000	2	NULL	NULL
8	Ashish	200	5000	2	NULL	NULL
1	Saurabh	900	12000	2	NULL	NULL

Question

Find 2nd highest salary in each department?

emp

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

Filtered 2nd highest salary of each dept by assigning dense rank to each record using window function dense rank.

```sql
SELECT *
FROM (SELECT emp.*,
DENSE_RANK() OVER(PARTITION BY department_id
ORDER BY salary DESC) AS rn
FROM emp) a
WHERE rn=2;
```

emp_id	emp_name	department_id	salary	manager_id	rn
1	Ankit	100	10000	4	2
3	Vikas	100	10000	4	2
7	Sanjay	200	9000	2	2

Question

Find records where

manager salary > employee salary?

emp

emp_id	emp_name	department_id	salary	manager_id
1	Ankit	100	10000	4
2	Mohit	100	15000	5
3	Vikas	100	10000	4
4	Rohit	100	5000	2
5	Udit	200	12000	6
6	Agam	200	12000	2
7	Sanjay	200	9000	2
8	Ashish	200	5000	2
1	Saurabh	900	12000	2

Filtered records where manager salary was greater than employee salary using SELF JOIN

```sql
SELECT e1.emp_id, e1.emp_name, e1.salary, e1.manager_id,
e2.emp_id, e2.emp_name, e2.salary, e2.manager_id
FROM emp e1
JOIN emp e2
ON e2.emp_id = e1.manager_id
WHERE e2.salary > e1.salary;
```

emp_id	emp_name	salary	manager_id	emp_id	emp_name	salary	manager_id
1	Saurabh	12000	2	2	Mohit	15000	5
8	Ashish	5000	2	2	Mohit	15000	5
7	Sanjay	9000	2	2	Mohit	15000	5
6	Agam	12000	2	2	Mohit	15000	5
4	Rohit	5000	2	2	Mohit	15000	5

Question

Update query to swap gender?

orders

customer_name	order_date	order_amount	customer_gender
Shilpa	2020-01-01	10000	Male
Rahul	2020-01-02	12000	Female
SHILPA	2020-01-02	12000	Male
Rohit	2020-01-03	15000	Female
Shilpa	2020-01-03	14000	Male

Updated customer gender one by one using Case Statement

```
UPDATE orders SET customer_gender = CASE
    WHEN customer_gender='Male' THEN 'Female'
    WHEN customer_gender='Female' THEN 'Male'
    END;
```

customer_name	order_date	order_amount	customer_gender
Shilpa	2020-01-01	10000	Female
Rahul	2020-01-02	12000	Male
SHILPA	2020-01-02	12000	Female
Rohit	2020-01-03	15000	Male
Shilpa	2020-01-03	14000	Female

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Thank You

